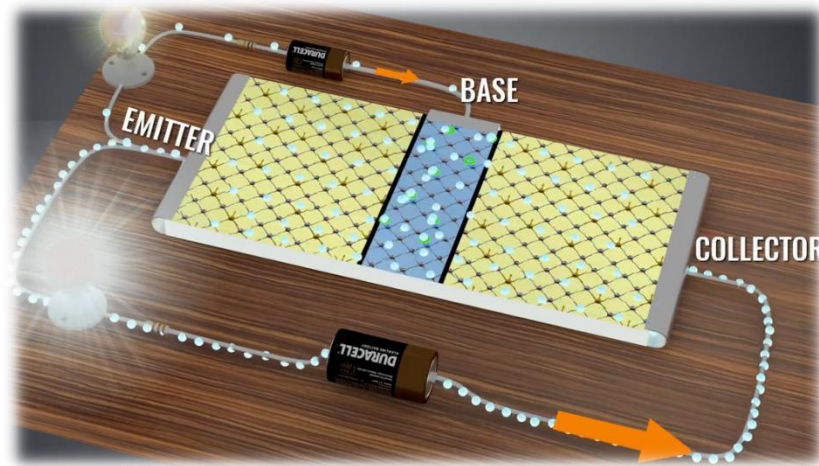


Analog Circuits

模拟电路



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Course Information

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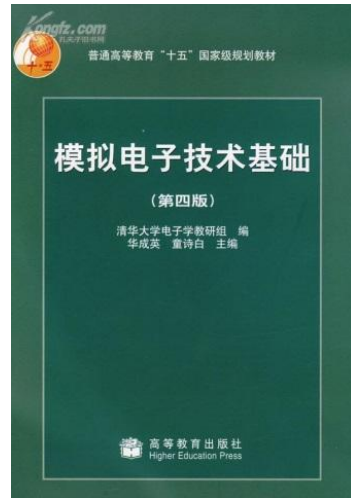
Class hours & Class rooms

14:00 – 15:50 (Tue.) , Room 506, 1st teaching building

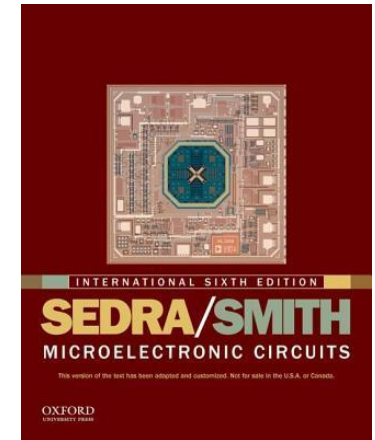
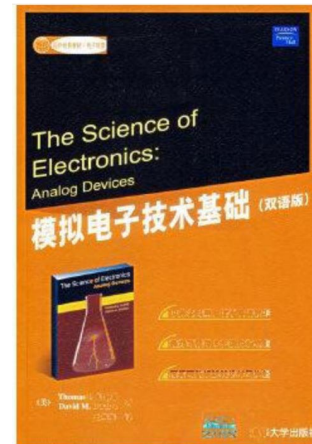
19:00 – 20:50 (Thur.) (1,3...15 week) , Room 506, 1st teaching building

Course Information

Chinese textbooks:



English textbooks:



Course Evaluation 考核方式

- ❑ Mid-term exam, 40%
- ❑ Final exam, 40%
- ❑ Homework, 20%

**Closed-book
exams**
闭卷考试

Total score: 100 points+ **3 points**

3 points: additional score for motivated students

Homework: 20 points

- ◆ 10 assignments in total 总共10次作业
- ◆ Publish on blackboard and QQ group
在blackboard和QQ group上发布
- ◆ Submit your homework on blackboard
需要在blackboard上提交电子版作业
- ◆ Delay of handing in homework will result in 20% deduction of your homework score
迟交作业会导致作业扣掉20%额外分数

Teaching language

100% English + 20% Chinese

80% English + 20% Chinese

60% English + 40% Chinese

40% English + 60% Chinese

:

0. Introduction

0.1 The development of electronics

0.2 Analog v.s. Digital

0.3 About this course

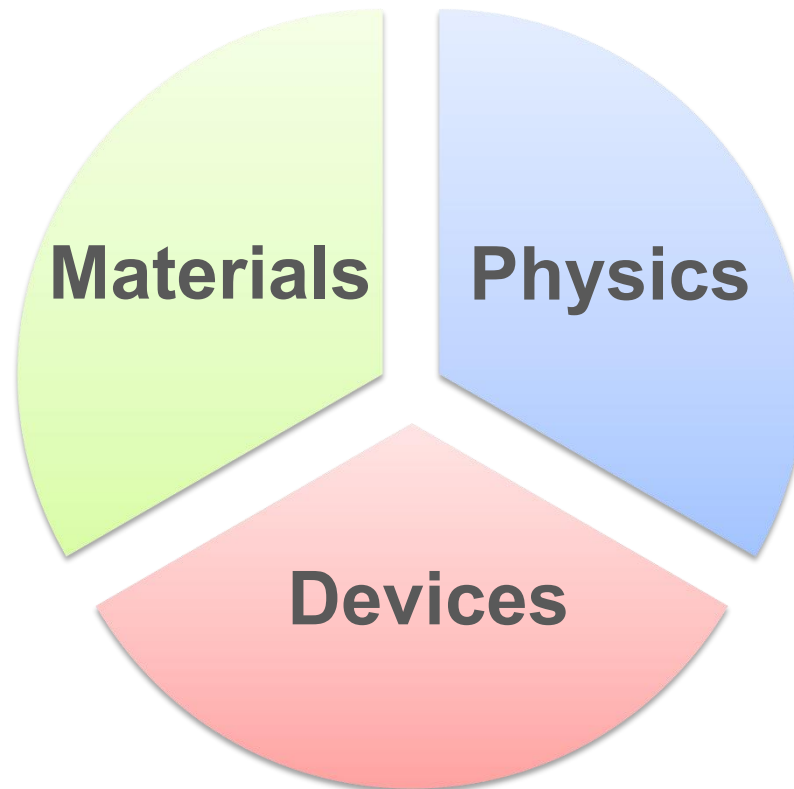
0.1 Electronics

Electronics?

Science and technology that focus on electronic devices, circuits and applications

电子技术就是研究电子器件、电子电路及其应用的科学技术。

The modern electronic science and technology is an interdiscipline (交叉学科) of materials, physics and devices.



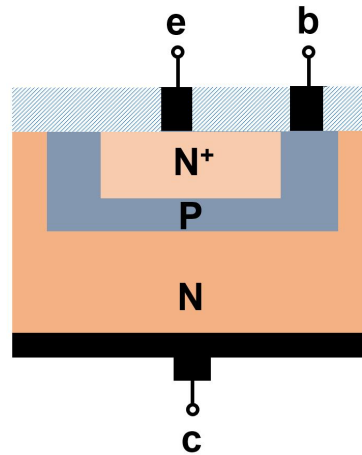
One example: the development of transistors

1904,
Vacuum tube



Heated cathode
emit electrons

1947,
Transistor (BJT)

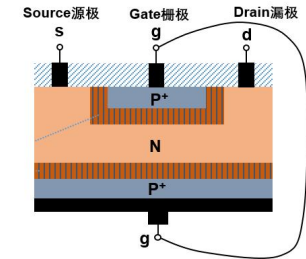


**Material: Silicon,
Germanium**

Device: N⁺PN. P⁺NP

Physics: PN junction

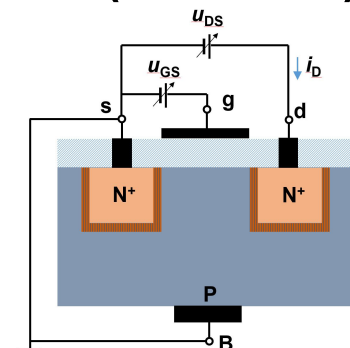
1950,
Junction field-effect
transistors (JFET)



New device design

Physics: field effect

1959,
Metal-oxide-semiconductor
FET (MOSFET)



Silicon is the king of the modern electronic world!



21th century: Electronic and Information World



Electronics changes our world and life in every aspects.

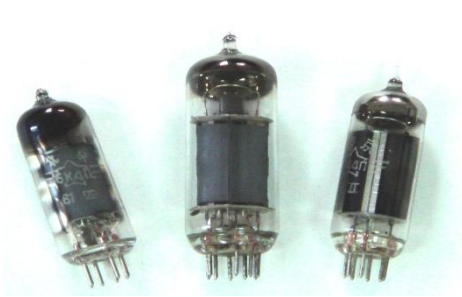
The evolution of human



Human = A pile of code in future?



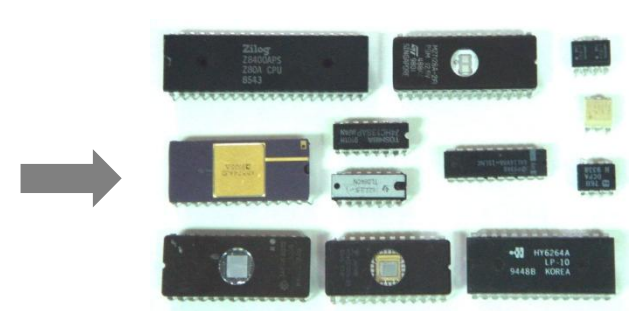
Development of electronic device



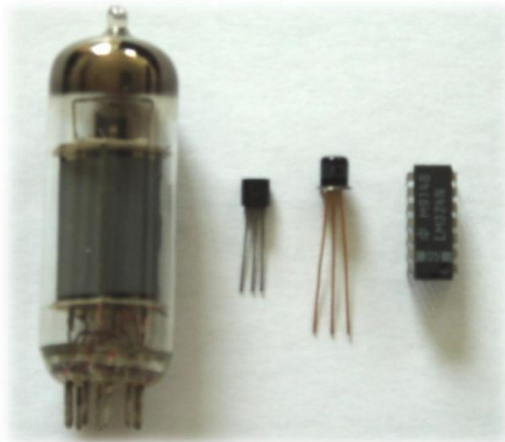
1904,
Electronic tube, 电子管



1947,
Transistor, 晶体管



1959,
Integrated circuit 集成电路



Comparison of electronic tube,
transistor, integrated circuit

电子管、晶体管、集成电路比较

Vacuum tube

1st generation electronic device 第一代电子器件

Electronic (vacuum) tube 电子管（真空管）



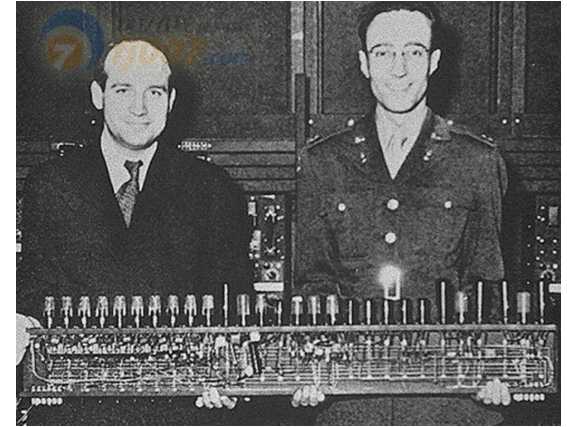
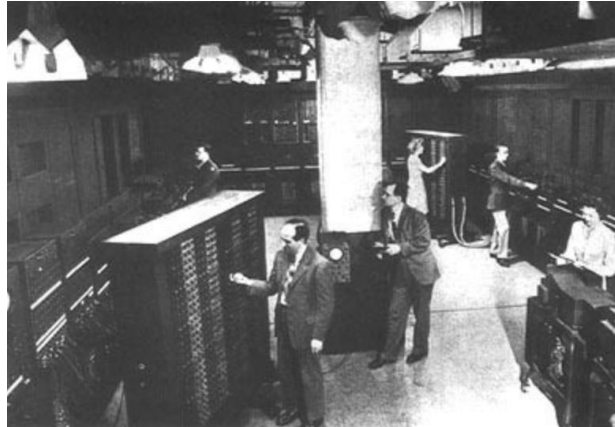
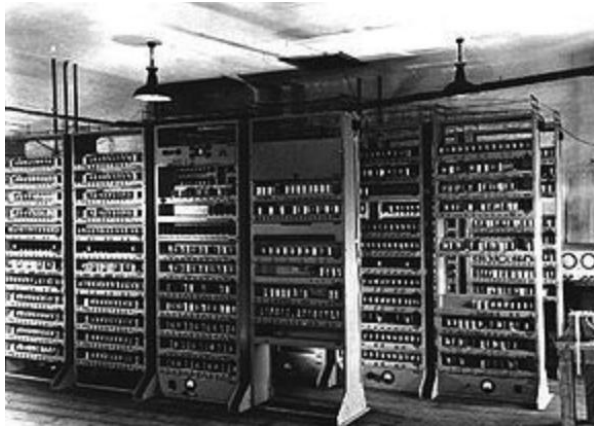
a. Vacuum shield

b. Cathode 阴极

c. Heated cathode will emit electrons 阴极可由灯丝加热，使温度升高，发射出电子

d. Current: Electrons motion under the electric field and magnetic field 电子受外加电场和磁场的作用下，在真空中运动就形成了电子管中的电流

1st computer is made of vacuum tube



The [ENIAC](#) was invented by [J. Presper Eckert](#) and [John Mauchly](#) at the University of Pennsylvania and began construction in [1943](#) and was not completed until [1946](#).

Length: 50 feet; Width: 30 feet; Area: 1500 feet²; Weight: 30 ton; Power consumption: 170 kW; 18000 vacuum tubes; Every 15min, one vacuum tube breaks; 据传 ENIAC 每次一开机，整个费城西区的电灯都为之黯然失色。

In one second, it processes 5000 times plus/minus calculation or 500 times multiplication calculation.

The disadvantage of vacuum tube:

Big, heavy, power consumption and short lifespan

体积大、重量重、耗电大、寿命短

The advantage of vacuum tube:

Can be applied in high power devices 高功率器件

Transistor

2nd generation electronic device-- **transistor** 晶体管

1948, Bell Lab



Transistor is made of **semiconductors**(半导体)

John Bardeen (1908-1991), Walter House Brattain(1902-1987), and William Shockley (1910-1989)

John Bardeen

Born in USA in 1908, attend university at 16, interest in Physics.

He won Nobel Physics Prize twice!

1956 for inventing transistors (Bardeen, Shockley, Brattain)

1972 for the BCS superconducting theory (Bardeen, Cooper, Schrieffer)

The legend of John Bardeen



约翰·巴丁(John Bardeen)

Will be introduced in
second chapter...

Transistor

Advantages:

- a. Small & light
- b. Long lifespan
- c. Low power consumption

Disadvantages:

- c. Sensitive to temperature
- d. Poor overload capacity

Electronic circuit 电子电路

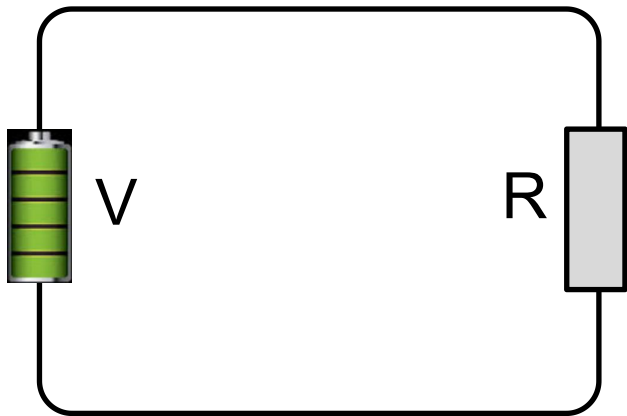
An electronic circuit is constructed through connecting electronic devices with resistors, inductors, capacitors, switches etc. to realize certain functions.

电子器件与电阻器、电感器、电容器、变压器、开关等元件适当地连接起来所组成的具有特定功能的电路。

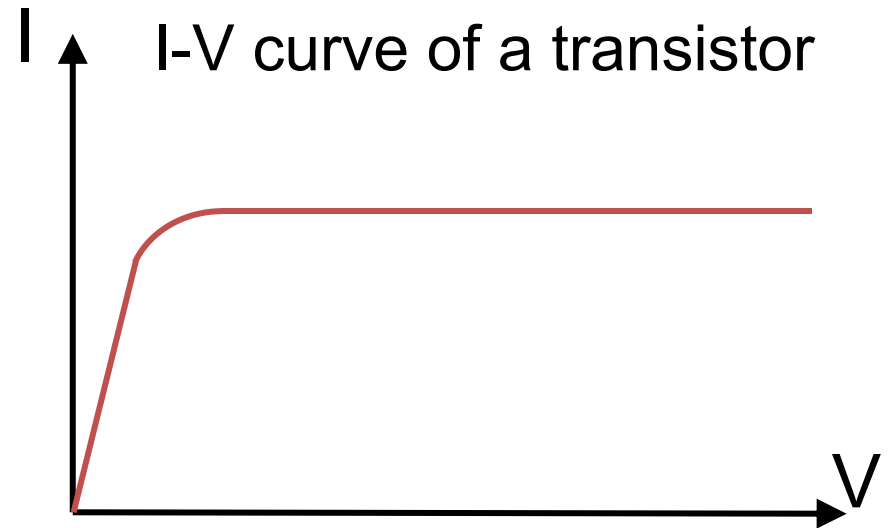
Electronic circuit

Difference compared with the ordinary circuit

- a. Electronic devices (diode, transistor)
- b. Non-linear characteristics

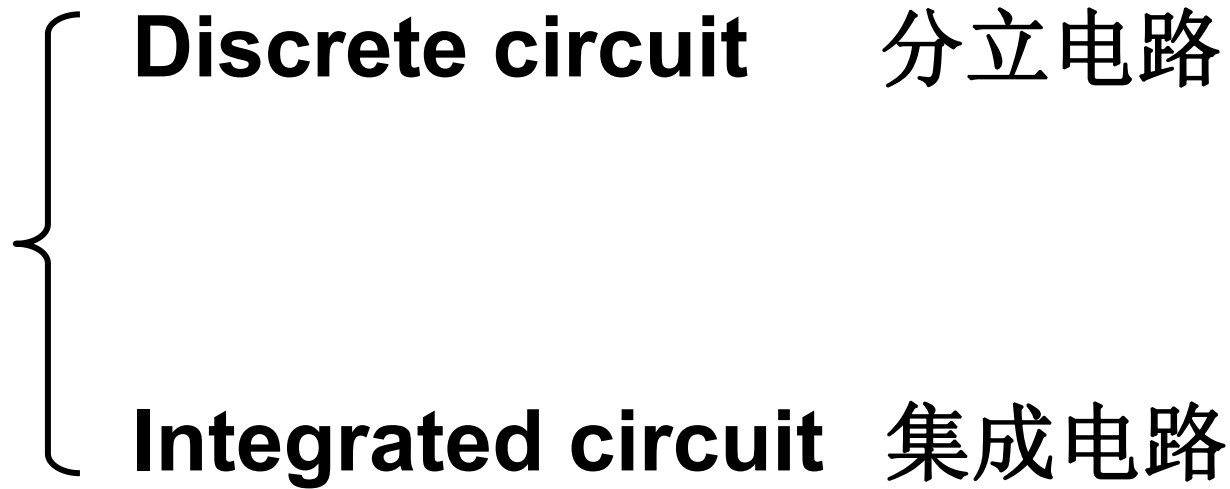


Linear circuit $I = \frac{V}{R}$



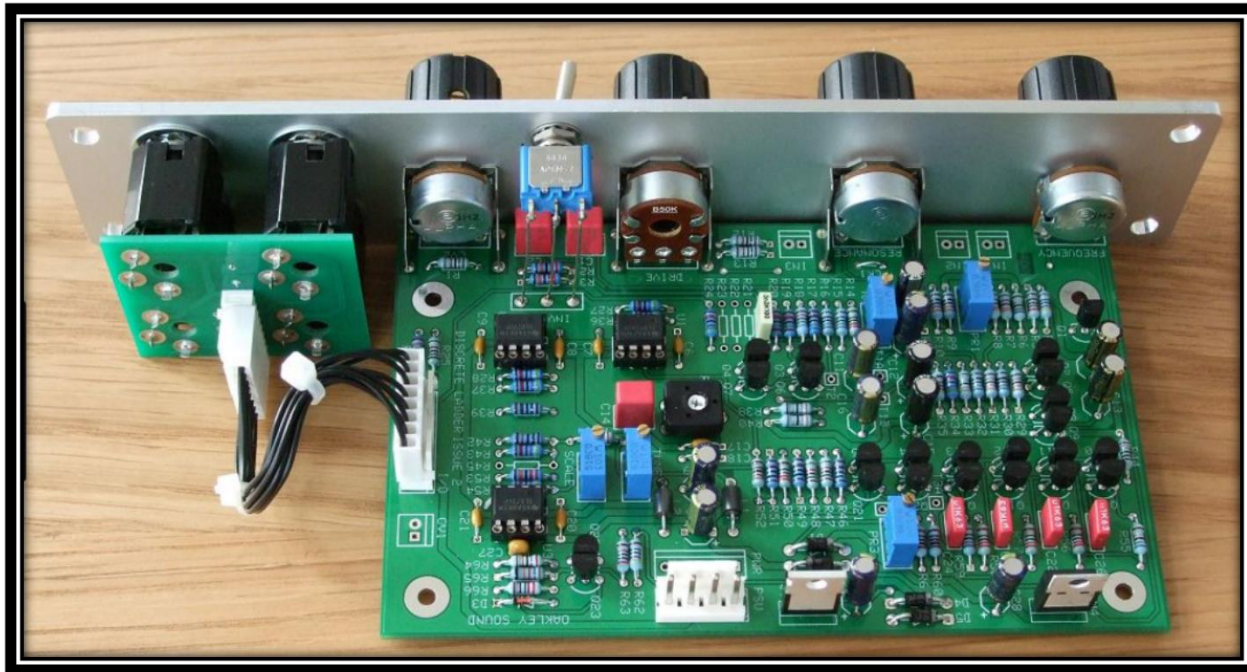
Non-linear

Electronic circuit



Discrete circuit

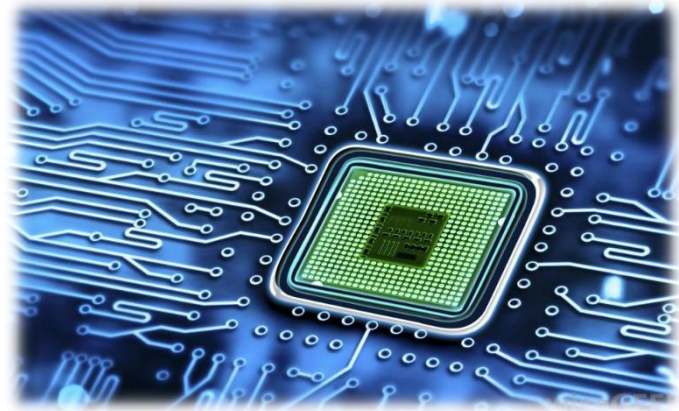
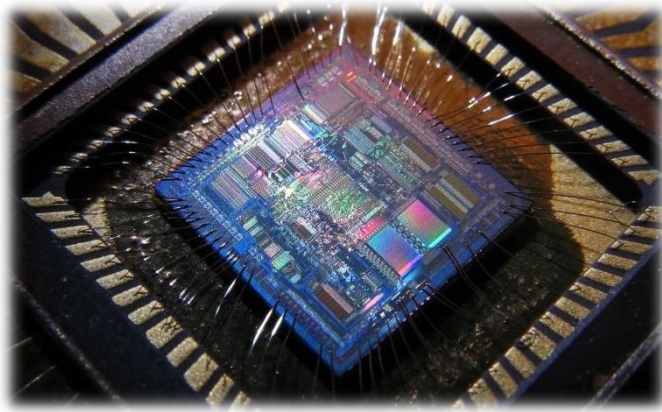
The electronic devices, resistors, capacitors and inductors are welded to the circuit board 把许多元件和器件焊接在印刷电路板上组成的



Large size; High power consumption; Low reliability

Integrated circuit-- IC

Integration of electronic devices (transistors) and components (resistors) on a single silicon chip



Advantages:

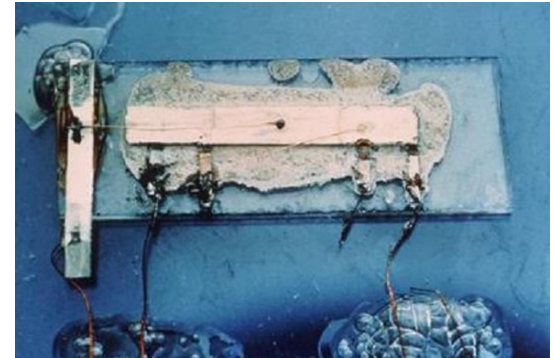
- a. Small & light
- b. Low power consumption
- c. Liability
- d. Long lifespan

Development of integrated circuit

The first IC is invented by [Jacky Kilby](#) in Texas Instruments in 1959

Only 4 transistors on germanium slab

Nobel prize in Physics, 2000



7/16 x 1/16 inch

(1) Small scale integration , SSI 小规模集成电路

Integration level: several tens of devices per chip

集成度：一块芯片上集成有几十个元器件。

Development of integrated circuit

(2) Medium scale integration , MSI 中规模集成电路

Integration level: several hundreds of devices per chip

(3) Large scale integration , LSI 大规模集成电路

Integration level: several thousands/tens-of-thousands of devices per chip

(3) Very large scale integration , VLSI

超大规模集成电路

Integration level: > several hundreds of thousands of devices per chip

Development of integrated circuit

(5) State-of-the-art of IC

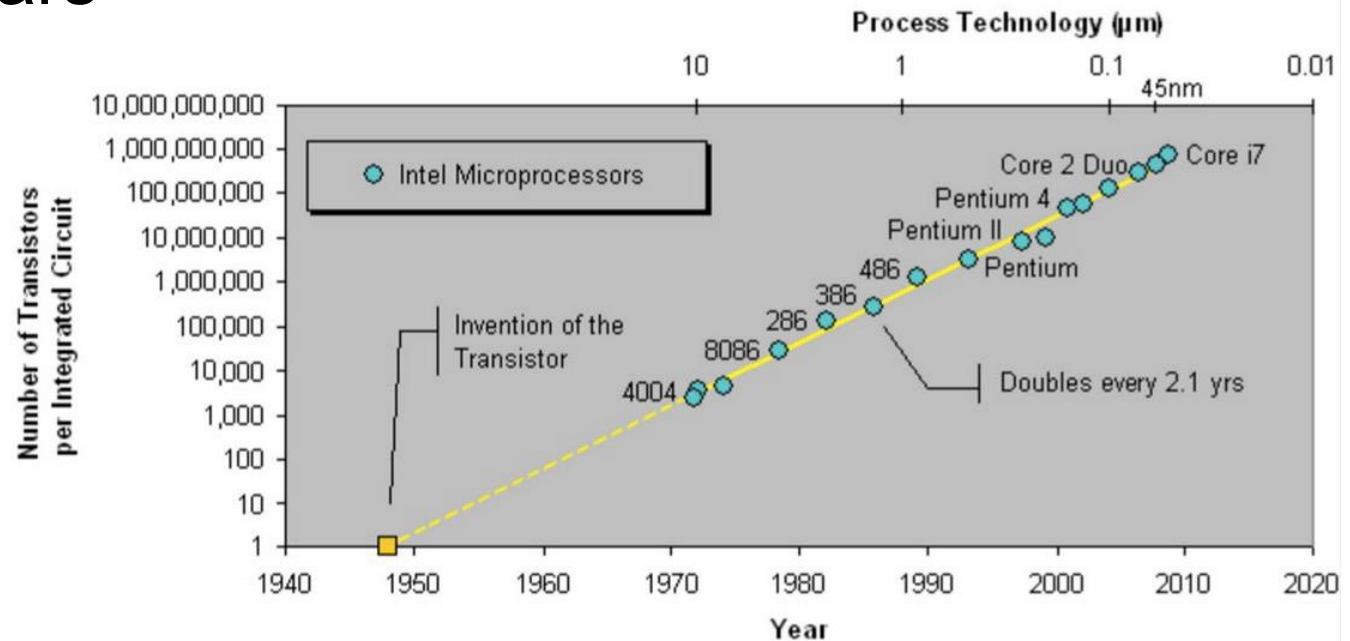
- a. Over one hundred million devices per silicon chip
- b. Very high speed and low power consumption



Let's see a video

Development of integrated circuit

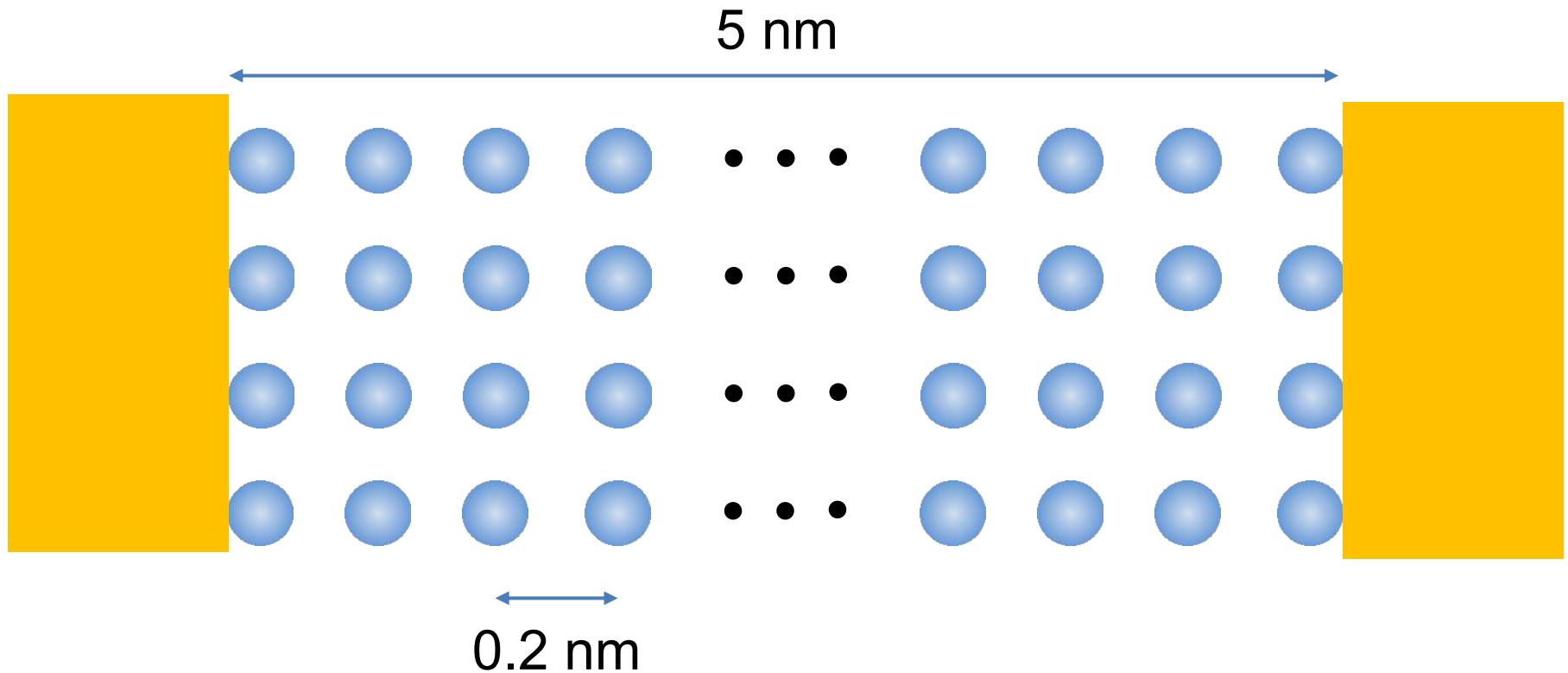
Moore's law: number of transistors per IC doubles every two years



Development is slower after 2013

State-of-the-art transistor length is ~ 5 nm: only several tens of atoms

A schematic of short-channel transistor



The channel length $\approx$ length of 25 atoms

0.2 Analog

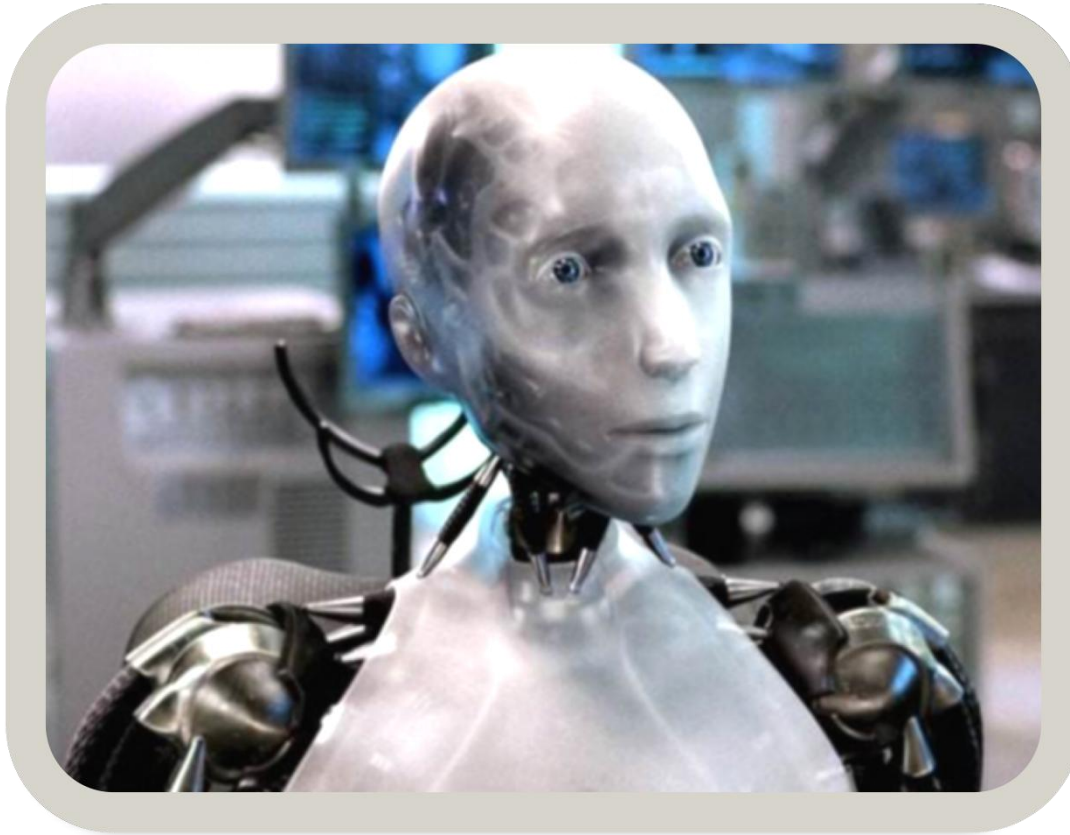
“Analogue” originates from French. It means “similarity”.

The word ‘analogue’ **was first used as a technical term during the 1940s**, and referred specifically to a class of computing technology.

Today, the word enjoys much wider usage, typically conveying **continuity**.

Signal 信号

Signal: what you see, what you hear, what you touch...



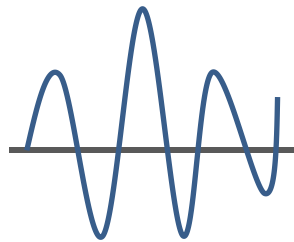
Q: Signal in nature is continuous or discrete?

Analog vs Digital

Category of electronic circuit based on **signal processing**

Analog模拟

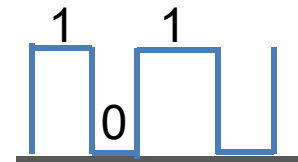
Signal is **continuous**



Device is in amplification mode

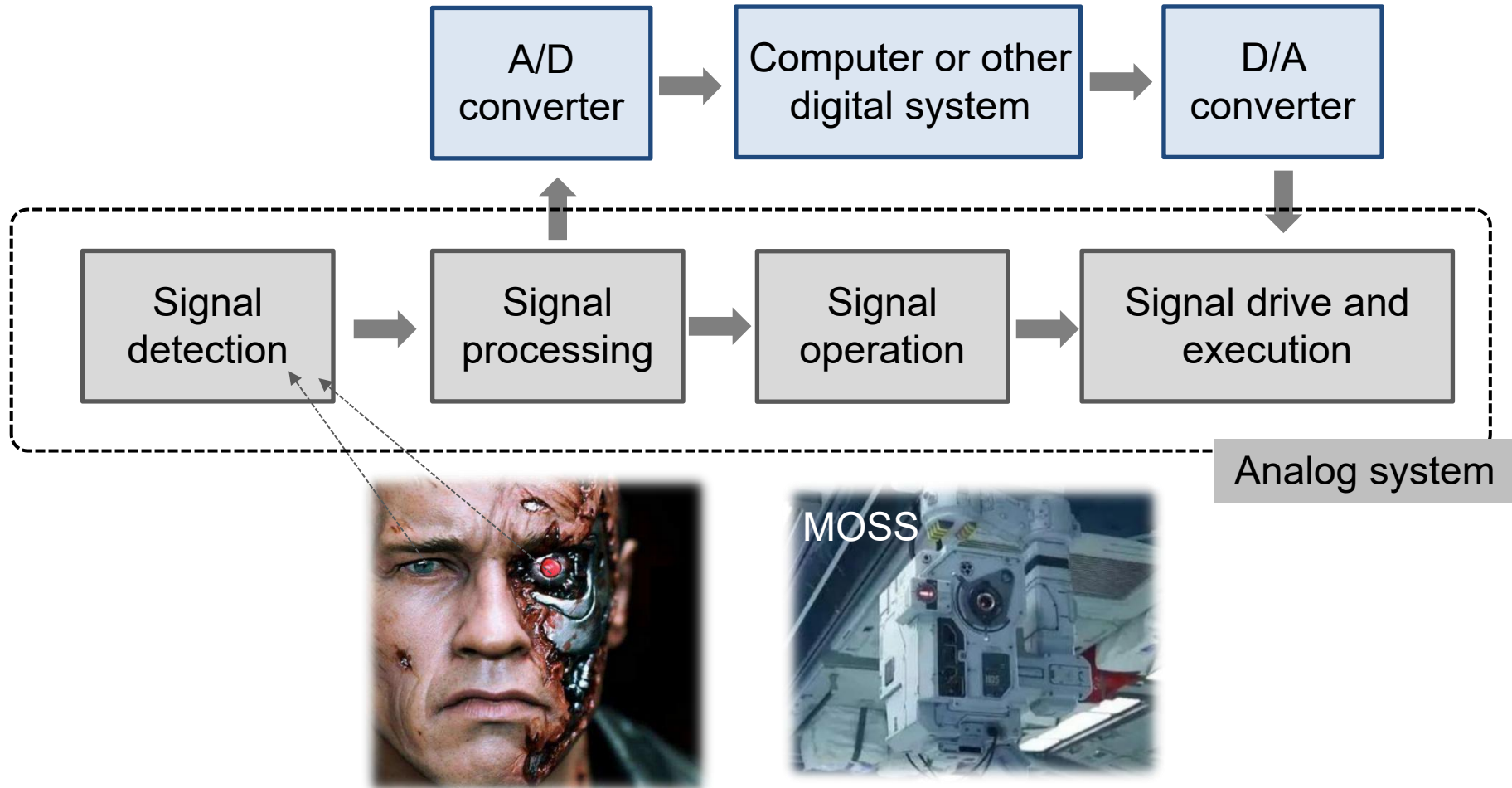
Digital数字

Signal is **discrete**



Device is in on/off mode

Electronic information system



About this course

Interesting



A lot of interesting devices and circuits.



Analog Circuit



Important



The core course for electronics.



Difficult



Too many concepts, devices, circuits, applications...

“四大名补” 之一

It's demon circuits!
魔鬼电路!



If you have enough will and determination, you can succeed!

